

AWS Aurora, for instance, is a distributed database but has limitations, such as a single writer node. Google Spanner is robust, but lacks MySQL compatibility, which is essential for many organisations. Microsoft Azure, on the other hand, does not yet have a full-fledged distributed OLTP (online transaction processing) database.

Q. What challenges can a customer face while adopting a distributed database?

A. Adopting a distributed database introduces complexities, particularly in operational management. Customers need advanced operational capabilities to monitor and manage their systems effectively. For instance, tracking CPU consumption, network bandwidth, query loads, and inter-module communication becomes critical. TiDB helps by providing robust tools and a detailed operational framework for monitoring these metrics.

Geo-distributed deployments present another challenge, especially when clusters are spread across different data centres. Ensuring network bandwidth and latency requirements are met is vital. TiDB addresses this by offering high availability and resilience. For example, when deployed on AWS, TiDB can replicate data across three availability zones. Even if an entire zone goes down, the database remains accessible, ensuring zero downtime.

Q. How is the open source nature of TiDB beneficial for its users?

A. Open source projects benefit from global community contributions, which drive innovation and rapid problem-solving. TiDB, as an open source solution, provides flexibility to users—they can try it for free, without any vendor involvement.

Due to the absence of vendor lock-in, TiDB can be deployed on AWS, Google Cloud, Microsoft Azure, or even

on-premises, depending on the user's needs, ensuring that organisations retain complete ownership of their data and database infrastructure.

Q. How do you see the adoption of open source databases globally, and how does it compare to that in India?

A. Open source adoption has been steadily increasing worldwide. Despite being open source, many organisations running mission-critical applications expect enterprise-grade support alongside it, which is understandable. However, the adoption of open source spans regions and industries. Digital-native companies—especially in sectors like e-commerce, logistics, and fintech—often prefer open source solutions due to their scalability and flexibility.

In India, the inclination towards open source is even stronger. Many Indian businesses have extensive IT teams with in-house expertise across various technologies. This makes them more self-reliant and capable of managing open source solutions independently. Open source fits well in such setups, allowing companies to rely on their internal teams instead of seeking external support. Opting for open source provides a sense of long-term security.

Q. Which industry segments are you targeting in India?

A. Our primary focus is on digital-native businesses, including e-commerce, logistics, fintech, gaming, and SaaS companies. These industries experience similar data growth patterns, requiring scalability, real-time analytics, and continuous expansion. Their needs align well with TiDB's strengths, making them our key target segments.

Q. Many consider open source solutions to be less secure and more vulnerable to attacks compared to proprietary options. How do you address these concerns?

A. Not all open source solutions are alike. They vary greatly depending on the development team and the work invested in the project. For TiDB, security has always been a top priority, especially since our clients work with core banking applications. We have implemented stringent security measures, including strong access controls. We comply with key security certifications such as ISO, PCI DSS (for financial data security), and HIPAA (for healthcare data security). Regular audits and certifications give us and our customers confidence in TiDB's security.

Q. In banking and credit card applications, where simultaneous transactions generate vast amounts of data, how does TiDB ensure continuity of data flow across distributed systems?

A. TiDB's distributed architecture is designed to ensure high availability, even in cases of infrastructure failure. Let me explain with a simplified example. If I have a record for my name, Bhanu, TiDB will store three copies of that record across different availability zones (AZs) within AWS. Even if one entire AZ goes down, the other two copies remain accessible, ensuring uninterrupted service.

This architecture allows TiDB to withstand infrastructure failures, whether on-premises or in the cloud. Even if a server or network experiences downtime, TiDB ensures data availability and continuity for end users. High availability is a cornerstone of TiDB's design, providing resilience and reliability in all scenarios.

Q. Is TiDB compatible with multi-cloud, hybrid cloud, and edge environments? If yes, how does it achieve this?

A. TiDB's open source nature ensures compatibility across a variety of environments. You can download